



Forecasting and Analyzing Predictors of Inflation Rate: Using Machine Learning Approach

Pijush Kanti Das¹ · Prabir Kumar Das¹

Accepted: 17 January 2024 / Published online: 29 February 2024
© The Author(s), under exclusive licence to The Indian Econometric Society 2024

Abstract

In this study, we investigate and apply the models from the machine learning (ML) paradigm to forecast the inflation rate. The models identified are ridge, lasso, elastic net, random forest, and artificial neural network. We carry out the analysis using a data set with 56 features of 132 monthly observations from January 2012 to December 2022. The random forest (RF) model can forecast the inflation rate with greater accuracy than other ML models. A comparison to benchmark econometric models like auto-regressive integrated moving average demonstrates the superior performance of the RF model. Moreover, nonlinear ML models are proven to be more successful than a linear ML or time series models and this is mostly due to the unpredictability and interactions of variables. It indicates that the significance of nonlinear structures for forecasting inflation is important. Furthermore, the ML models outweigh the benchmark econometric model in forecasting the undulations due to the COVID-19 impact. The findings in this study support the benefit of applying ML models to forecast the inflation rate. Even without considering the sporadicity of pandemic, nonlinear model like artificial neural network (ANN) outweighs other models. Additionally, the ML models like RF and ANN model yield variable importance measures for each explanatory variable. ML models shows capability to not only better forecasting but also able to provide the insight regarding the covariates for improved forecasting results and policy prescriptions.

Keywords Inflation · Forecasting · Random forest · Artificial neural networks · Machine learning

JEL Classification C4 · C5 · C8

✉ Pijush Kanti Das
pijush_phdmp19@iift.edu; pkdas.stat@gmail.com

¹ Indian Institute of Foreign Trade, 1583, Chowbaga Canal Side Rd, Madurdaha, Chowbaga, Kolkata, West Bengal 700107, India

Introduction

Forecasting inflation is one of the most researched topics including the recent developments in big data and statistical learning techniques (deep and machine learning) (Goulet Coulombe et al. 2022). The motivations of this study are to explore the increasing adoption of machine learning (ML) models to elucidate complex phenomena of macroeconomic indicator such as inflation rates. One primary assumption behind this motivation driving the shift towards ML models possibly lies in their capacity to capture complex, non-linear relationships inherent in economic data unlike traditional econometric models, which often assume linear relationships. In the realm of inflation dynamics, where multifaceted factors interplay, the flexibility of machine learning might become invaluable. Machine learning models, equipped with the ability to adapt and self-adjust to evolving environments, outshine econometric models that may struggle to accommodate sudden shifts in economic conditions such as geopolitical changes, technological advancements, or global economic crises etc. Also, machine learning models demonstrate a superior ability to handle high-dimensional data, extracting meaningful insights from vast datasets that may overwhelm traditional econometric approaches. This might play a crucial role in the context of inflation rate modeling, where numerous variables contribute to the overall economic picture. Econometric models often rely on strict assumptions regarding the nature of relationships between variables. Machine learning models, on the other hand, operate with fewer a priori assumptions, allowing them to discover and adapt to patterns organically. This reduces the risk of model misspecification and enhances the robustness of the model in capturing the true dynamics of inflation.

Predicting inflation rates using machine learning models is an interesting economic problem as well as a technical one. The main aspect of our research is the monetary policy by the policymakers, economist and businesses, where inflation forecasts play a very important role. The production of accurate and reliable inflation forecasts is necessary and important to enable monetary authorities to adequately deal with inflationary pressures that may arise in the future (Binner et al. 2005). Economic systems are highly complex and dynamic, influenced by a myriad of factors including known factors like interest rates, unemployment rates, money supply, government policies, global economic conditions etc. and also to many unknown factors. Also, economic relationships are often non-linear and non-monotonic. Traditional economic models may struggle to capture these complexities, whereas machine learning models, particularly those capable of handling non-linear relationships, may offer more flexibility in representing the underlying patterns in the data (Mullainathan and Spiess 2017; Goulet Coulombe et al. 2022). Inflation expectations have also long been thought to play a central role in monetary policy and economic analysis in many countries, including India, at least since President Milton Friedman's 1968 speech at the American Economic Association (Hall and Sargent 2018). Unfortunately, it is impossible to know exactly what people's inflation expectations are. People have different ideas about what inflation is and how high it can get (Patra and Ray 2010). Several

studies also show that survey responses regarding inflation expectations exhibit high volatility (Blanchflower and MacCoille 2009; Goyal and Parab 2019). The interest of this research was to explore the realm of machine learning models to obtain more accurate and near real time forecasts of inflation, using India, as a case study.

Review of Studies

Traditional econometric approaches in forecasting time series are based on Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA) (Box et al. 2015), ARIMA with explanatory variables (ARIMAX) (Liu 1986) and Vector Auto Regression (VAR) (Sims 1980). Modern econometric thinking for forecasting inflation may be started with the improvement of Phillips curve-based estimation, using diffusion indexes and the Unobserved Component Stochastic Volatility (UCSV) model (Stock and Watson 1999, 2002). As Stock and Watson (2007) continued their study for the US, many scholars devoted their time to developing improved forecast measures across the globe (Brouwer and Ericsson (1998) for Australia; Binner et al. 2005 for the Euro area). Other than traditional approaches, several univariate and multivariate models have been studied for forecasting inflation rates. However, such models have mostly based on the linearity assumption and do not account for the uncertainty of macroeconomic variables. Linear models usually have poor forecasting performance due to the data's inherent non-linearity (Binner et al. 2005). These models are restricted by many limitations and assumptions. With the development of machine learning (ML) and more notably deep learning (DL) algorithms, forecasting has become more data-driven with the capability to fit both nonlinear parametric as well as nonparametric models incorporating many features.

Zhang et al. (1998) explored the promise of artificial neural networks (ANN) and raised several future research questions in terms of improvement over traditional time series models, choice of the best algorithm for a particular forecasting problem, and designing of data. In quest of this, several scholars have used several machine learning techniques on macroeconomic data and concluded a better forecasting performance over a benchmark model (in most cases Random Walk (RW) or ARIMA). The algorithms have been studied for both uni-variate and multivariate setups. On the other hand, after the development of Bootstrap aggregating (Bagging) (Breiman 1996) and Random Forest Model (Breiman 2001) along with other ML techniques like Ridge Regression (RR) (Hoerl and Kennard 1970), Least Absolute Shrinkage and Selection Operator (LASSO) (Tibshirani 1996), Adaptive Lasso (AdaLASSO) (Zou 2006), ElasticNet (ElNet) (Zou and Hastie 2005), forecasting performance has improved.

A vast body of literature has already been contributed to assess the efficiency of ML Techniques, some of which may only be discussed as a reference here. Inoue and Kilian (2008) observed Bagging provides more accurate forecasting measures for US CPI-based inflation over median forecasts, shrinkage estimators, and Bayesian model averaging.

Nakamura (2005) used uni-variate neural network models to forecast inflation and showed its efficiency over benchmark AR models. Lately, Medeiros et al. (2021), forecasted inflation using ML methods in a data-rich environment. Medeiros et al. (2021) considered RW, AR, and UCSV models as a benchmark and compared them with shrinkage models like RR, LASSO, AdaLASSO, and ElNet; Factor models like target factors and factor boosting; Ensembled methods like Bagging, Complete Subset Regression, Random Forests (RF), Jackknife Model Averaging, and Hybrid Models like (RF/OLS) and (adaLASSO/RF). A data-rich environment has been created in terms of a sample (672 observations, January 1960–December 2015) and variables. In this analysis, 122 variables and four lags of all variables were used, for a total of 508 potential predictors. All variables have monthly frequencies and are transformed in year-to-year growth. The models were compared in terms of forecasting performance using root mean square error (RMSE), mean absolute error (MAE), and mean absolute deviation (MAD). Medeiros et al. (2021) concluded that ML methods (LASSO, RF) can produce better forecasts than benchmarks. RF models deserve special attention due to their having the smallest forecasting errors owing to their nonparametric nature and their variable selection mechanism. RF also came out effective as a forecasting method for both economic recession and expansion periods and hence in both low and high uncertainty periods, which is a significant property for any macroeconomic variable.

In India, the Reserve Bank of India (RBI) conducts an inflation expectations survey of households (IESH) quarterly to estimate three months ahead and 1-year ahead of inflation expectations. However, Sharma and Bicchil (2018) showed that survey respondents do not build expectations rationally, hinting that the IESH by the Reserve Bank of India (RBI) has low credibility. Das et al. (2019) analyzed the dynamics of IESH data and concluded quantification of directional expectations is better than the simple average of the quantitative expectations on the future inflation rate.

The literature on machine learning methods for inflation forecasting for India is scarce, though some recent ones may be worth mentioning. Pratap and Sengupta (2019) estimated CPI inflation with several machine-learning models using uni-variate data and observed that machine-learning models can generally outperform standard econometric models. Pradhan (2011) used ANN models in his study and observed that simple multivariate ANN models forecast better over the other uni-variate models. Thakur et al. (2016) analyzed the monthly data of India from January 2000 to December 2012 and built an inflation forecasting model based on a neural network model. Rani et al. (2017) outlined forecasting of monthly inflation using ANN in a multivariate setup. However, none of these studies evaluated the effectiveness of ML models with a large number of variables in the context of developing countries like India. Furthermore, the impact of a pandemic such as COVID-19 on forecasting macroeconomic parameters like inflation has not been studied before for any country.

The objective of this work is to fill this gap and prepare a user friendly database for predicting inflation in Indian context, since India is a typical example of a developing economy. The objective of this study is to answer the question whether there are viable ML models with multiple features that can provide improved accuracy in

forecasting the inflation rate. The study also compares benchmark time series methods to ML models to determine how well they perform in uncertain situations like COVID-19.

Methodology

Benchmark Model

We used non seasonal auto regressive integrated moving average (ARIMA) model as our benchmark model. The ARIMA model is a widely used time series forecasting method that combines the elements of autoregression, differencing, and moving averages. The model can be written as:

$$y_t' = c + \phi_1 y_{t-1}' + \dots + \phi_p y_{t-p}' + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t, \quad (1)$$

where, y_t' is the the autoregressive component involves modeling the relationship between an observation and several lagged observations (previous time steps). The “p” parameter represents the number of lag observations included in the model. A higher “p” value indicates a longer memory of past values. The integrated component involves differencing the time series data to make it stationary. In this ARIMA(p,d,q) model, the differencing parameter “d” represents the number of times differencing is applied to achieve stationarity. If the original data is not stationary, differencing helps remove trends and seasonality. The moving average component models the relationship between an observation and a residual error from a moving average model applied to lagged observations. The “q” parameter represents the size of the moving average window. A higher “q” value indicates a longer memory of past forecast errors. The best ARIMA(p,d,q) model was chosen based on the automatic selection process of the program based on the optimization of Bayesian Information Criterion (BIC).

Regularized Regression

Regularized regression is a type of linear regression that includes a regularization term in the objective function. This regularization term is added to the ordinary least squares (OLS) objective to avoid overfitting and enhance the model’s generalization performance. The two most common forms of regularized regression are Ridge Regression (L2 regularization) and Lasso Regression (L1 regularization).

Ridge Regression (L2 Regularization)

Ridge regression adds a penalty term to the sum of squared coefficients in the OLS objective function (James et al. 2013). The objective of Ridge Regression is to minimize the following:

$$\text{Objective} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2, \quad (2)$$

where, n is the number of observations, p is the number of features or explanatory variables, y_i is the observed response for the i th observation, $\hat{y}_i$ is the predicted response for the i th observation, β_j is the coefficient for the j th feature and λ is the regularization parameter.

The term $\lambda \sum_{j=1}^p \beta_j^2$ penalizes the model for large coefficients. As λ increases, the impact of this penalty becomes stronger, leading to smaller coefficients and a simpler model.

Lasso Regression (L1 Regularization)

The LASSO (Least Absolute Shrinkage and Selection Operator) regression, originally proposed by Tibshirani (1996), similar to Ridge, adds a penalty term to the OLS objective, but it uses the absolute values of the coefficients. The objective of Lasso Regression is to minimize the following:

$$\text{Objective} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j|, \quad (3)$$

where, all terms are the same as in Ridge Regression in Eq. 3. The term $\lambda \sum_{j=1}^p |\beta_j|$ introduces sparsity into the model. It tends to force some of the coefficients to be exactly zero, effectively performing feature selection.

Elastic Net (EINet) Regression

Elastic Net Regression combines both L1 and L2 regularization terms in the objective function. It includes both the penalties for large coefficients (L2) and the sparsity-inducing penalties (L1) (Zou and Hastie 2005).

$$\text{Objective} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \left(\rho \sum_{j=1}^p |\beta_j| + \frac{1-\rho}{2} \sum_{j=1}^p \beta_j^2 \right), \quad (4)$$

where, ρ is the mixing parameter, with $0 \leq \rho \leq 1$, when $\rho = 1$, it is equivalent to Lasso Regression and when $\rho = 0$, it is equivalent to Ridge Regression.

Regularized regression techniques are widely used in machine learning and statistics to build models that generalize well to new data while handling multicollinearity and preventing overfitting.

Random Forest (RF)

Breiman (1996) proposed the RF model, which is an ensemble learning method for making predictions that combines multiple decision trees. Each decision tree is trained on a random subset of the data, and the final prediction is made by averaging (for regression) or voting (for classification) the predictions of individual trees.

The following steps are being followed to run the regression through RF model in this study,

- For each tree, the RF algorithm recursively split the data into subsets based on feature values.
- At each node, the algorithm selected the feature that best separates the data into homogenous groups (minimizing the variance of the target variable within each group).
- The splitting process continues until a predefined stopping criterion is met based on the minimum size of the parent node to limit tree growth (Breiman 2001).
- The prediction for a new data point is the average of the target variable values in the leaf node to which it belongs.

Mathematically, the prediction for a new data point, x , in a regression tree can be represented as:

$$\hat{y}(x) = \frac{1}{n} \sum_{i=1}^n y_i, \quad (5)$$

where, n is the number of samples in the leaf node to which x belongs, and y_i is the target variable value for each sample in that leaf node. In this study randomforest package in R has been used.

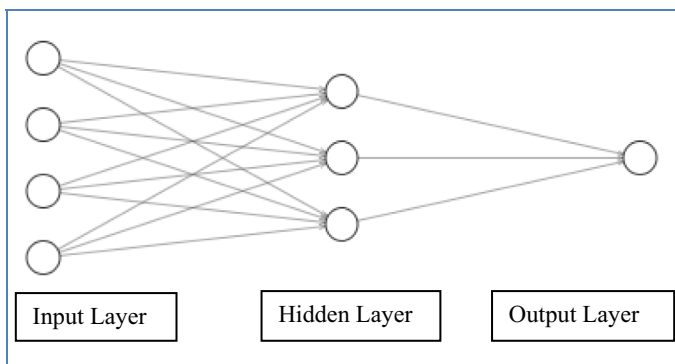


Fig. 1 ANN architectural diagram (Source: Authors' drawing)

Artificial Neural Network (ANN) Model

An Artificial Neural Network (ANN) is a machine learning model inspired by the structure and functioning of the human brain. It consists of interconnected nodes, or artificial neurons, organized into layers. The three main types of layers in a typical ANN are the input layer, hidden layer, and output layer (Fig. 1).

The number of nodes in the input layer corresponds to the number of explanatory variables in the input dataset. Each node in a hidden layer performs a weighted sum of the inputs and applies an activation function to produce an output. The weights are parameters that the network learns during the training process. The output layer produces the final results of the network's computation. The connections between nodes in different layers have associated weights. These weights are adjusted during the training process to minimize the difference between the predicted output and the actual target values. Activation functions introduce non-linearity to the network, enabling it to learn complex patterns and relationships in the data. The loss function measures the difference between the predicted output and the actual target values. The goal during training is to minimize this loss.

The equations for an artificial neural network (ANN) involve the computation within each node, which includes the weighted sum of inputs, the application of an activation function, and the propagation of the output to subsequent layers. Here are the key equations for a simple feedforward neural network used in this study with one hidden layer:

Weighted Sum for a Node (Including Bias)

$$Z_j = \sum_{i=1}^n w_{ij}x_i + b_j, \quad (6)$$

where, Z_j , is the weighted sum for node j in the current layer, w_{ij} is the weight connecting the i th node in the previous layer to the j th node in the current layer, x_i is the output of the i th node in the previous layer and b_j is the bias for node j .

Activation Function: $a_j = f(z_j)$, where, a_j is the output of the activation function for node j and $f(\cdot)$ is the activation function, such as the sigmoid function, hyperbolic tangent (tanh), or Rectified Linear Unit (ReLU).

Output of the Network: $\hat{y} = f(z_k)$, where, $\hat{y}$ is the predicted output of the network, $f(\cdot)$ is the activation function in the output layer and z_k is the weighted sum for the output node.

Loss Function: $\text{Loss} = L(\hat{y}, y)$, where, $L(\cdot)$ is the loss function that measures the difference between the predicted output ($\hat{y}$) and the actual target (y).

Model Evaluation

In order to evaluate the forecasting performance of each model, we relied on Root Mean Squared Error (RMSE) and relative RMSE. Model has been evaluated for both trained and test data. As the model was already built based on trained data, it is more important to observe the prediction accuracy in respect of test data.

$$\text{RootMeanSquaredError(RMSE)} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (7)$$

$$\text{RelativeRMSE}_i = \frac{\text{RMSE of Model}_i}{\text{RMSE of Benchmark model}} \quad (8)$$

Variable Importance in Random Forest Model

Earlier ML models were treated with little scope on the insights about the explanatory variable. But, recent developments in ML algorithms surpassed this barrier.

Variable importance in a Random Forest model is a measure of the impact or contribution of each feature (variable) in predicting the target variable. Random Forest models are an ensemble of decision trees, and they assign importance scores to each feature based on how much they contribute to reducing the impurity or error in the predictions.

A variable importance measure based on the difference in accuracy of predictions arising from different permutations of trees was developed by Breiman et al. (1984) and Breiman (1996). We describe the variable importance here. Let β^t represent the number of out of bag samples for a tree t and let $L(P_t(x_i), y_i)$ represent the prediction accuracy at the i th training sample. Equation 9 shows the importance (significance) of variable X_j in tree t .

$$VI^{(t)}(X_j) = \sum_{i \in \beta^t} L(P_t(x_i), y_i) - L(P_t(x_{i,\pi_j}), y_i), \quad (9)$$

where, $x_{i,\pi_j} = (x_{i,1}, \dots, x_{i,\pi_j(i)}, x_{i,j+1}, \dots, x_{i,p})$ and where π_j is a random permutation of n integers. In classification settings the prediction accuracy $L(P_t(x_i), y_i)$ is defined as

$$L(P_t(x_i), y_i) = \frac{\sum_{i \in \beta^t} I(\hat{y}_i^t = y_i)}{|\beta^t|}, \quad (10)$$

where, $\hat{y}_i^t = P_t(x_i)$ denotes the prediction made by tree t at point x_i , and $I(\cdot)$ denotes the indicator function. The variable importance measure for variable X_j is determined by adding the importance of all trees in the forest.

$$VI(X_j) = \frac{\sum_{t \in B} VI^{(t)}(X_j)}{n_{tree}}, \quad (11)$$

Garson's Algorithm for Variable Importance in ANN Model

The major objective of this study to handling more new variables through machine learning model in understanding inflation and get some value addition in traditional economics concepts.

In order to do so, the relative importance of explanatory variables in relation to inflation rate has to be determined.

The relative importance of an explanatory variable is the amount that each variable contributes to the prediction of the target variable. One approach, first proposed by Garson (1991) and later refined by Goh (1995), is to deconstruct model weights to determine the relative importance of a specific explanatory variable within a supervised neural network. The relative importance (or strength of association) was determined by taking into account all weights associated with a specific input node that passes through the hidden layer to the target variable. The same process is repeated for all explanatory variables. The connections are then compared with each other for all input nodes. Finally, a single value for each explanatory variable is obtained that provides the relationship with target variable. The absolute magnitude of relative importance of the variables was scaled between 0 and 1.

$$P_{ij} = |W_{ij}| * |W_{i1}|, \quad (12)$$

where, W_{ij} is the weight of the input node j in hidden layer i and W_{i1} is the weight of the output node 1 in the hidden layer i . Second, for each hidden layer i , P_{ij} is divided by the sum of all P_{ij} s for the input variables to calculate Q_{ij} as shown in Eq. 13. The first subscript represents the hidden layer number, the second subscript represents the input node number and m represents the total number of inputs.

$$Q_{ij} = P_{ij} / (P_{i1} + P_{i2} + P_{i3} + \dots + P_{im}), \quad (13)$$

At that point, for each input variable j , S_j is calculated utilizing Eq. 14.

$$S_j = Q_{1j} + Q_{2j} + Q_{3j} + \dots + Q_{kj}, \quad (14)$$

where, k is the number of hidden layers. Then, for each input node j , to calculate Relative Importance (RI) of node j , we divide S_j by the sum of all S_j values and express it as a ratio, as shown in Eq. 15

$$RI_j = S_j \times 100 / (S_1 + S_2 + S_3 + \dots + S_m), j = 1(1)m, \quad (15)$$

where, m is the number of input variables. Therefore, variables with higher RI have higher importance in predicting the target variable, i.e. inflation rate.

Data Sources and Preprocessing

Data Sources

We used feature variables and inflation rate data from India for this investigation. In India, there are four price indices that are released: the RBI's Wholesale Price Index (WPI); the Labor Bureau's Consumer Price Index (CPI) for three different worker categories (industrial, urban non-manual, and agricultural/rural sectors); and the Ministry of Statistics and Programme Implementation's (MOSPI) Gross Domestic Product (GDP) deflator. The WPI and CPI are published monthly, with a one month lag for the provisional index and three months lag for the final index. WPI inflation was the official inflation rate for India until 2014. Since 2014, it was decided to calculate CPI based Index for calculation of Inflation. Most countries use the CPI to assess inflation because it measures changes of the cost of living at consumers end.

Several research studies are conducted to explore the reasons for inflation in various countries, and a vast number of influencing variables have been found. These determinants have been classified as supply-side and demand-side factors in economic terms (Gordon 1975). From the economic perspective, these determinants have been categorized to supply-side and demand-side factors. Supply-side factors are those economic factors that cause inflation by increasing cost of the production (cost-push inflation). Some important supply-side factors include output growth, capital formation, oil and import prices, tax and wage levels, and exchange rate. On the other hand, demand-side factors lead to higher inflation via creating more buying requests for goods and services in the country (demand pull information). Some important demand side factors are money growth, private consumption expenditure and government consumption expenditure etc. considering this, we have observed a vast literature to consider all determinants and total 56 variables have been considered. Justification of undertaking all variables are being skipped here for time being as it would take a lot of additional space. However, metadata has been prepared and the list of all explanatory variables with references and relationship with inflation as per available literature has been provided in Annexure 1. This analysis of metadata further helped in considering maximum number of important variables in this study.

Data of all the variables in this study are extracted from the CMIE database (CMIE 2022). For CPI, the data is available from January 2011 to December 2022. On the other hand, WPI is available since April, 1982. In this paper, we focus on the study of the inflation rate based on two periods of CPI, one is the whole set i.e., January, 2012–December, 2022 and another to March, 2020 to observe the effect of COVID-19 situation in India after March, 2020. It may be mentioned here, the data has been compiled by CMIE from different Official data sources like MoSPI, Government of India and RBI and we have cross checked the data with original sources.

Preprocessing

According to Raj et al. (2008), the standard year-on-year inflation rate is more reliable and significant method of analyzing inflation than the monthly inflation rate. A smoothed and less volatile annual inflation rate would be more beneficial than a highly volatile monthly inflation rate from the standpoint of policy objectives like price stability. To calculate Inflation rate, in this study we have used year-on-year inflation. e.g., to calculate Inflation Rate as on March, 2020, we observed the changes in CPI on March, 2020 from March, 2019. Also, for all the explanatory variables we used the year-on-year growth to nullify the effect of stationarity.

The frequency of data points is another important aspect of this analysis. We have considered the variables only with data availability of “monthly” frequency for the time being. Another important aspect of the preparation of data is to make it spliced/linking series in terms of base. All relevant data for the different base periods has been corrected to make a single series data. For example, CPI data has been made a single series data with base 2012 = 100.

To sum up, our data consists of 132 time points/observations (from January 2012 to December 2022) and consists of 56 variables (including Inflation Rate) which describe the Indian Economy and probable predictors for Inflation Rate. For using ML techniques, we divided our data set into two parts, namely, analysis sample (January 2012–August 2019; 92 observations) as trained data sample on which the models are built, and test sample (September 2019–December 2022; 40 observations) as test data, where the models are tested. The partitioning of the data set is based on 70% of training set and 30% of test splitting scheme (Siarni-Namini et al. 2019).

Also, the data up to March, 2020 has been used in this study to study the negative effect of the COVID—19 pandemic in Indian Economy where lock down started since April, 2020. There, our data consisted of 99 time points/observations for each variable (from January, 2012 to March, 2020) and the data again

Table 1 RMSE and relative RMSE for both train and test data for all models including COVID-19 period (January, 2012–December, 2022)

Model name	RMSE_Trained	RMSE_Test	RelativeRMSE Trained	Relative RMSE_ Test
Benchmark: ARIMA	1.841	2.885	1.000	1.000
Ridge regression	0.622	55.208	0.338	19.134
LASSO regression	0.671	11.540	0.364	3.999
ElNet regression	1.371	2.148	0.745	0.744
Random forest	0.360	1.654	0.196	0.573
Artificial neural network	0.644	2.961	0.350	1.026

Source: Authors' calculation

divided into two periods, i.e. January 2012–December 2017 (72 observations) as trained data and January 2018–March 2020 (27 observations) as test data. The data analysis is conducted in R software using neuralnet, fastshap, boot and plyrpackages.

We have used min–max normalization technique for scaling of the data. We have used k fold cross validation technique to remove bias and increase robustness and tested the model based on RMSE value in each sample.

Empirical Results

Descriptive Statistics

Table 4 in Annexure-A reports the summary statistics for Inflation Rate and all 56 explanatory variables from January, 2012 to December, 2022. A standard deviation of 2.3 for Inflation rate and standard deviation of many other explanatory variables suggests that Inflation and other explanatory variables for India are less persistent and more volatile. The histogram of the variables also shows that many of the variables are highly skewed.

Main Results

First, after multicollinearity test using variance inflation factor (VIF), we have kept only 42 explanatory variables after removing the highly correlated variables to run for all our models. In Table 1, we report the value of RMSE and Relative RMSE for different models under study with respect to the benchmark auto regressive moving average (ARIMA) model for both train and test data.

Table 1 shows, RF model outperforms all other forecasting models to predict inflation rate (relative RMSE for test data=0.013). It may also be observed that

Table 2 RMSE and Relative RMSE for both train and test data for all models before COVID-19 period (January, 2012–March, 2020)

Model Name	RMSE_Trained	RMSE_Test	RelativeRMSE_Trained	Relative RMSE Test
Benchmark: ARIMA	0.593	1.789	1.000	1.000
Ridge regression	0.884	2.413	1.491	1.349
LASSO regression	0.886	2.370	1.495	1.325
ElNet regression	0.324	2.390	0.546	1.336
Random forest	0.956	2.882	1.611	1.612
ANN	0.543	1.637	0.916	0.915

Source: Authors' calculation

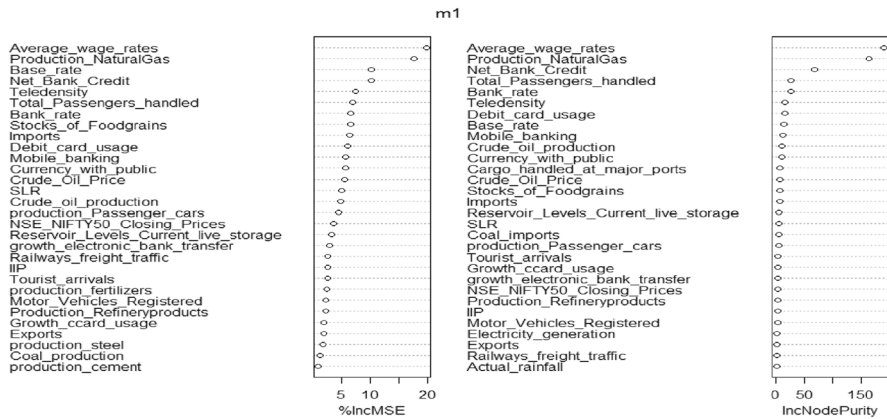


Fig. 2 Variable importance plot for random forest model including COVID-19 period. Source: Author's calculation

from RMSE test value, other ML Models like ElNet regression (2.148), ANN Model (2.961) and Random Forest model (1.654) fits way better than benchmark ARIMA Model (2.885). It may clearly be seen Random Forest model provides best forecasting performance with a relative RMSE of 0.573 among the considered models. Table 2 summarizes the analysis with the data only before COVID-19 period (January, 2012–March, 2020). First thing, we can observe here the RMSE value of test data in Table 2 is much lower than the RMSE value of test data in Table 1, specially,

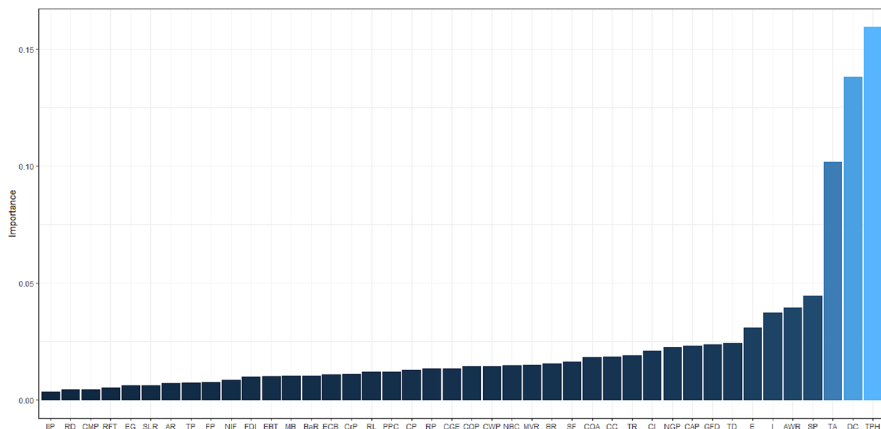


Fig. 3 Variable importance in ANN model up to COVID-19 period, Source: Authors' calculation

for the regularized regression models. The higher RMSE in Table 1 is due to the fluctuation during COVID-19 pandemic. It has been also observed that in before COVID-19 data, ANN model outperforms all other models. Also, ARIMA provides better forecasting than Random Forest and other regularized regression, when there is no undulation in the data.

From both the tables, it is clear that Machine Learning models are performing better than benchmark econometric methods and capture the non-linearity and sporadicity of the data. The random forest model may outperform others as it is non-parametric in nature and due to its variable selection mechanism based on classification helped in better forecasting. This supports the results of Medeiros et al. (2021) and Masini et al. (2023) even in Indian context. Non-linear ML models like RF and ANN combine with large data sets may direct the Indian economic prediction to a new high.

The variable importance in random forest model for the data including COVID-19 period (January, 2012–December 2022) is depicted in Fig. 2. In Fig. 2, the first part provides the mean decrease of accuracy in predictions (%IncMSE) and the 2nd part the total decrease in node impurity (IncNodePurity). In both cases, higher value shows the higher importance of variables. It just means e.g., if average wage rate variable is excluded the MSE will be increased by 20%. Most important variables from both part are found to be are: average wage rate, natural gas production, bank and base rate, net bank credit, tele-density, debit card usage, mobile banking, currency with public etc.

Figure 3 illustrates the relative importance of each explanatory variable for explaining inflation rate from the ANN model. The variable names in abbreviation in the graph may be seen in Table 3 of Annexure 1. The results suggest that number of passengers handled in aviation sector, debit card usage, tourist arrivals, steel production, average wage rate, exports, imports, tele-density etc. are the most influential variables for predicting inflation rate from ANN model.

Conclusion

This paper forecasts inflation rate using CMIE's large-scale macroeconomic database with more than fifty variables. Also, we present a non-exhaustive review of the machine learning models to forecast macroeconomic indicator. The output of the analysis throughout the research under this paper is useful in various ways. The machine learning models excelled in processing and analyzing the data with large number of explanatory variables and provided smaller forecasting error than the traditional time series model. Even, out of different machine learning models the results showed the applicability of neural network and random forest model as per

different situation. The feature importance algorithms in ANN and RF model uncovered subtle patterns and relationships that may not be apparent through traditional statistical methods and identified the relevant variables that contribute the most to predict inflation. Further, based on the results of this research, models can be trained to provide real-time or near real-time predictions, which is crucial for policymakers and businesses who need timely information to make informed decisions in a rapidly changing economic environment.

Salient Findings

Some important findings have been explored from the study.

First, ML models, viz. RF and ANN models outperform all other models. In sporadic situations like COVID-19 pandemic, even Random Forest (RF) model performs better than ANN model.

Second, all machine learning models, viz.- Shrinkage models (Lasso Regression, Ridge Regression and Elastic Net Regression), Random Forest (RF) and Artificial Neural Network(ANN) model showed better performance in forecasting inflation rate than benchmark auto regressive moving average model (ARIMA).

Third, RMSE of Random Forest model had almost 43% improvement over benchmark ARIMA model for the dataset including COVID-19 period and ANN had almost 9% improvement over benchmark model excluding the COVID-19 data. It may be observed as COVID-19 period has more sporadicity in data is better estimated by RF model on the other hand ANN model without COVID-19 period showed better performance.

Further, the feature or variable importance in ML models has been studied for better understanding of the influence of explanatory variables on inflation. Even though ML models are able to handle large number of explanatory variables, feature importance for both RF and ANN model can highlight the most influential variables on Inflation rate. Average wage Rate, net bank credit, tele-density, debit card usage, passengers handled in aviation etc. might play more influential role in inflation targeting.

Implications

Predicting macroeconomic indicators with greater precision was always a big interest to the economists, forecasters and policymakers. This study clearly shows

that using of ML models have more gain than traditional econometric methods in predicting inflation rate. This could pave the way for more accurate forecasting of other major macroeconomic indicators such as gross domestic product (GDP), unemployment rate etc. At present, estimating these indicators through official statistics takes significant times, but application of ML models on high dimension data might play a crucial role in timely use of these important indicators for policy uses in future.

Way Forward

Further, this study may be extended using latest ML models like long-short term memory networks (LSTM), transfer model etc. or by ensembling of different ML models to gain in forecasting performance. The results may also be studied for different forecasting horizons.

While machine learning models offer advantages in predicting inflation, it's important to note that they are not without challenges. Ensuring the reliability of predictions and understanding the limitations of the models are essential aspects of using machine learning in economic forecasting. Additionally, economic expertise is still crucial for interpreting results and incorporating domain knowledge into the modeling process. The results from this research suggest that, despite being limited by a lack of large data samples in macroeconomics, Random Forest and NN can be successfully apply in the field. The field of ML algorithm development is an ongoing process; however, we hope that the observations presented here will be useful to anyone interested in applying ML techniques to macroeconomic forecasting, particularly in the context of a developing country.

Annexure A

Table 3 Selection of explanatory variables based on literature

Sl. no.	Name of variables	Source of literature	Observed relationship with inflation rate as per literature
1	Inflation rate (based on CPI) (Inf)	Target variable	–
2	Currency with Public (CWP)	Milo, P. (2012). The impact of the budget deficit on the currency and inflation in the transition economies. <i>Journal of Central Banking Theory and Practice</i> , 1(1), 25–57	Positive
3	M1 growth (M1)	Jane M Binner*, Rakesh K Bissoondeal, Thomas Elger, Alicia M Gazely, and Andrew W Mullineux. A comparison of linear forecasting models and neural networks: an application to euro inflation and euro divisia. <i>Applied Economics</i> , 37(6):665–680, 2005	Positive
4	Net Bank Credit growth (NBC)	Korkmaz, S. (2015). Impact of bank credits on economic growth and inflation. <i>Journal of Applied Finance and Banking</i> , 5(1), 51	No relation
5	Cash reserve ratio (CRR)	Okotori, T. W. (2019). The Dynamics of monetary policy and inflation in Nigeria. <i>IOSR Journal of Economics and Finance (IOSR-JEF)</i> , 10(2), 37–49	Positive
6	Statutory liquidity ratio (SLR)	Chaudhary, N. Impact Of Inflation On Monetary Policy Of Indian Economy. <i>International Journal of Research in Management & Social Science</i> , 50	Positive
7	Bank rate (BR)	Koç, Ü., & Şahin, H. (2021). The role of electronic payments in inflation dynamics. <i>International Journal of Economic & Administrative Studies</i> , (32)	Negative
8	Electronic Payment Transfer (EBT)		Positive
9	Mobile banking (MB)	Geanakoplos, J., & Dubey, P. (2010). Credit cards and inflation. <i>Games and Economic Behavior</i> , 70(2), 325–353	Positive
10	Credit card usage (CC)		Positive
11	Debit card usage (DC)	Ashima Goyal, Prashant Parab, et al. Modeling heterogeneity and rationality of inflation expectations across Indian households. Indira Gandhi Institute of Development Research, Working Paper, 2019	Negative
12	Base rate (BaR)		Negative
13	Crude oil production (CrP)		
14	Tax receipt (TR)	Poterba, J. M., & Rotemberg, J. J. (1988). Inflation and taxation with optimizing governments	Both positive and negative

Table 3 (continued)

Sl. no.	Name of variables	Source of literature	Observed relationship with inflation rate as per literature
15	Central government expenditure (CGE)	Mehrara, M., & Sujoudi, A. (2015). The relationship between money, government spending and inflation in the Iranian economy. <i>International Letters of Social and Humanistic Sciences</i> , 51, 89–94	No significant effect
16	Revenue deficit (RD)	Ezebasili, V. N., Mojekwu, J. N., & Herbert, W. E. (2012). An empirical analysis of fiscal deficits and inflation in Nigeria. <i>International Business and Management</i> , 4(1), 105–120	Positive
17	Gross fiscal deficit (GFD)	Sahoo, Malayaranjan, and Narayan Sethi. "The dynamic relationship between export, import and inflation: Empirical evidence from India." <i>The Indian Economic Journal</i> 66.3–4 (2018): 294–311	Positive
18	Exports (E)	Onaran, O., & Obst, T. (2016). Wage-led growth in the EU15 member-states: the effects of income distribution on growth, investment, trade balance and inflation. <i>Cambridge Journal of Economics</i> , 40(6), 1517–1551	Positive
19	Imports (I)		
20	Trade Balance (TB)		
21	Net foreign direct investment (FDI)	Omankhanlen, A. E. (2011). The effect of exchange rate and inflation on foreign direct investment and its relationship with economic growth in Nigeria	No effect
22	ECCBs and FCCBs_total (ECB)	Misra, P., & Shettigar, J. (2020). Macroeconomic Variables Affecting External Commercial Borrowings: An Investigation. In <i>Innovation, Technology, and Market Ecosystems</i> (pp. 241–263). Palgrave Macmillan, Cham	Positive
23	Tourist arrivals (TA)	Shaari, M. S., Ahmad, T. S. T., & Razali, R. (2018). Tourism led-inflation: A case of Malaysia. In <i>MATEC Web of Conferences</i> (Vol. 150, p. 06026). EDP Sciences	Positive
24	Total foreign exchange reserves (TFER)	Gour Sundar Mitra Thakur, Rupak Bhattacharyya, and Seema Sarkar Mondal. Artificial neural network based model for forecasting of inflation in India. <i>Fuzzy Information and Engineering</i> , 8(1):87–100, 2016	Negative
25	Exchange rate (ER)		Negative

Table 3 (continued)

Sl. no.	Name of variables	Source of literature	Observed relationship with inflation rate as per literature
26	Actual rainfall (AR)	Dua, P., & Goel, D. (2021). Determinants of inflation in India. <i>The Journal of Developing Areas</i> , 55(2)	Positive
27	Coal imports (CI)	Guo, J., Zheng, X., & Chen, Z. M. (2016). How does coal price drive up inflation? Reexamining the relationship between coal price and general price level in China. <i>Energy Economics</i> , 57, 265–276	Negative
28	Coal production (COA)	Philippe Goulet Coulombe, Maxime Leroux, Dalibor Stevanovic, and St'ephane Surprenant. How is machine learning useful for macro-economic forecasting? arXiv preprint arXiv:2008.12477, 2020	Positive
29	Production_Natural gas (NGP)		Positive
30	Production_Refinery (RP) products		Positive
31	Production_fertilizers (FP)		Positive
32	Production_steel (SP)		Positive
33	Production_cement (CP)		Positive
34	Production_electricity (EP)		Positive
35	Production_Tea (TP)		Positive
36	Index of industrial production (IIP)		Positive
37	IIP mining (IIPM)		Positive
38	IIP manufacturing (IIPMa)		Positive
39	IIP electricity (IIPe)		Positive
40	Production_Passenger cars (PPC)		Positive
41	Electricity generation (Utilities) (EG)	Atems, B., & Hotaling, C. (2018). The effect of renewable and non-renewable electricity generation on economic growth. <i>Energy Policy</i> , 112, 111–118	Positive
42	Railways: freight traffic (RFT)	Mandal, S., & Chakraborty, S. <i>International Journal of Business Economics And Management Research</i>	Positive

Table 3 (continued)

Sl. no.	Name of variables	Source of literature	Observed relationship with inflation rate as per literature
43	Cargo handled at major ports (CMP)	Cheteni, P. (2013). Transport infrastructure investment and transport sector productivity on economic growth in South Africa (1975–2011). <i>Mediterranean Journal of Social Sciences</i> , 4(13), 761	Positive
44	cargo handled at aviation (CHA)	Alici, A., & Akar, A. S. (2020). Macroeconomic determinants of air cargo demand: a panel data analysis. <i>International Journal of Shipping and Transport Logistics</i> , 20(48), 11–23	Negative
45	Capital issues (CAP)	Olanipekun, B. O., & Akeju, F. K. (2013). Money supply, inflation and capital accumulation in Nigeria. <i>Journal of Economics and Sustainable Development</i> , 4(4)	Positive
46	Average wage rates for rural labourers in agricultural and non-agricultural sectors (AWR)	James H Stock and Mark W Watson. A comparison of linear and non-linear univariate models for forecasting macroeconomic time series. Technical report, National Bureau of Economic Research, 1998	Positive
47	Total: aircraft movements (TAM)	Aderamo, A. J. (2010). Demand for air transport in Nigeria. <i>Journal of Economics</i> , 1(1), 23–31	Negative
48	Total: passengers handled (TPH)		Negative
49	NSE_NIFTY50_Closing prices (NIF)	Atsushi Inoue and Lutz Kilian. How useful is bagging in forecasting economic time series? A case study of us consumer price inflation. <i>Journal of the American Statistical Association</i> , 103(482):511–522, 2008	Positive
50	Crude oil (petroleum) price (COP)		Positive
51	Bank credit to commercial sector and components (BCCSC)	Korkmaz, S. (2015). Impact of bank credits on economic growth and inflation. <i>Journal of Applied Finance and Banking</i> , 5(1), 51	No relationship found
52	Stocks of foodgrains (SF)	KU, G., & Pandit, V. (2014). Production, procurement and inflation: A market model for foodgrains. <i>Margin: The Journal of Applied Economic Research</i> , 8(4), 431–456	Positive
53	Reservoir levels: current live storage (RL)	Rains, Reservoirs and food prices Live Mint. https://www.livemint.com/Opinion/OvuM9z8ErSWAufCqCTXsdN/Rains-reservoirs-and-food-prices.html	Positive

Table 3 (continued)

Sl. no.	Name of variables	Source of literature	Observed relationship with inflation rate as per literature
54	Motor vehicles registered (MVR)	Soaring used car prices are pushing inflation higher, and there's not much the U.S. can do about it. CNBC. https://www.cnbc.com/2022/01/13/why-used-car-prices-are-pushing-inflation-higher.html	Positive
55	Mobile subscribers (MS)	Ehsan, Z. A. (2021). Assessing the nexus between mobile financial service usage and inflation—evidence from Bangladesh	Positive
56	Tele-density (TD)	Adeleye, N., & Eboagu, C. (2019). Evaluation of ICT development and economic growth in Africa. <i>NETNOMICS: Economic Research and Electronic Networking</i> , 20(1), 3	Positive

Table 4 Descriptive statistics

Variable Name	.Mean	SD	Min	Median	Max	Histogram
Inflation_Rate	5.9	2.3	1.5	5.6	11.5	
Currency_with_public	12.7	17.5	-47.6	11.6	105.7	
M1	11.7	7.8	-18.9	11.3	44.7	
Variable Name	.Mean	SD	Min	Median	Max	Histogram
Net_Bank_Credit	11.9	9.4	3.6	10.2	92.5	
CRR	-1.7	15.1	-29.2	0.0	33.3	
SLR	-2.5	2.1	-6.5	-2.5	0.0	
Bank_rate	0.9	21.6	-32.0	-5.6	58.3	
Base_rate	-1.2	4.8	-10.3	-1.6	21.6	
Growth_electronic_bank_transfer	16.1	10.2	-7.9	16.1	55.8	
Growth_ccard_usage	34.1	19.7	-3.4	30.8	184.4	
Debit_card_usage	106.1	546.5	-39.2	15.3	3962.3	
Mobile_banking	186.5	117.3	-29.4	186.5	472.3	
Tax_receipt	14.8	23.7	-44.3	13.0	151.8	
Central_government_expenditure	13.5	27.4	-36.9	12.3	212.0	
Revenue_deficit	-39.5	549.1	-5974.1	-1.2	966.2	
Gross_fiscal_deficit	32.0	311.7	-1273.0	9.5	2739.7	
Exports	5.9	24.8	-60.9	1.8	202.3	
Imports	7.4	28.8	-59.7	2.9	169.5	
Trade balance: DGCI&S	8.9	150.6	-1453.6	5.3	659.4	
Net_FDI	89.2	551.6	-3116.2	9.0	4301.0	
ECBs&FCBs_total	31.0	108.2	-86.7	-2.4	522.0	
Tourist_arrivals	86.3	334.6	-99.6	7.2	2691.4	
Total_foreign_exchange_reserves	6.2	9.7	-17.3	6.3	27.4	
Exchange_rate	-4.5	5.6	-20.0	-4.7	7.0	
Actual_rainfall	20.2	81.0	-89.2	1.7	537.9	
Coal_imports	11.8	26.4	-48.8	8.7	101.9	
Coal_production	4.5	8.5	-20.4	3.8	33.5	
Crude_oil_production	-2.4	2.6	-7.4	-2.1	5.6	
Production_NaturalGas	-4.4	8.2	-19.9	-4.1	25.0	
Production_Refineryproducts	2.2	7.7	-24.2	2.2	30.9	
Production_fertilizers	1.2	6.0	-14.3	1.2	18.0	
Production_steel	9.0	44.4	-82.8	5.5	472.7	
Production_cement	9.0	52.7	-85.2	4.3	582.7	
Production_electricity	6.0	7.6	-22.9	5.6	38.5	
Electricity_generation	4.8	7.1	-25.8	4.6	42.5	
Production_Tea	2.9	17.3	-53.6	3.0	90.3	
Railways_freight_traffic	4.6	9.3	-35.3	4.1	70.8	
Cargo_handled_at_major_ports	3.0	7.3	-23.2	2.8	31.6	
cargo handled at aviation	7.2	43.3	-82.5	5.3	445.4	
IIP	3.7	14.1	-57.6	3.4	133.5	
IIP_Mining & quarrying	1.8	8.2	-26.9	2.2	36.6	
IIP_Manufacturing	4.1	19.2	-66.6	3.2	196.0	
Variable Name	.Mean	SD	Min	Median	Max	Histogram
IIP_Electricity	6.0	7.3	-22.9	5.5	38.5	

Table 4 (continued)

Production_Passenger_cars	45.9	514.7	-98.6	-0.4	5927.5	
Capital_issues	25.2	68.4	-73.2	9.4	269.5	
Average_wage_rates	9.3	8.5	2.2	5.2	39.7	
Total_Aircraft_movements	29.2	222.5	-97.3	8.2	2516.8	
Total_Passengers_handled	172.0	1744.7	-99.8	13.0	20046.3	
NSE_NIFTY50_Closing Prices	12.6	17.2	-26.0	10.4	70.9	
Bank_Credit_to_CommercialSector	10.6	3.8	4.2	10.0	19.2	
Stocks_of_Foodgrains	3.4	17.7	-37.2	6.9	31.0	
Reservoir_Levels_Current_live_storage	5.8	27.9	-45.1	-0.6	122.8	
Motor_Vehicles_Registered	7.8	34.4	-88.1	5.2	225.4	
Mobile_Subscribers	2.8	5.3	-6.5	2.2	16.1	
Teledensity	1.5	5.3	-7.6	1.0	14.6	
Crude_Oil_Price	9.2	40.5	-71.1	7.3	208.1	

Data availability Data can be made available on request by the corresponding author.

References

- Binner, J.M., R.K. Bissoondeeal, T. Elger, A.M. Gazely, and A.W. Mullineux. 2005. A comparison of linear forecasting models and neural networks: An application to Euro inflation and Euro Divisia. *Applied Economics* 37 (6): 665–680.
- Blanchflower, D.G., and C. MacCoille. 2009. *The formation of inflation expectations: an empirical analysis for the UK* (No. w15388). National Bureau of Economic Research.
- Box, G.E., G.M. Jenkins, G.C. Reinsel, and G.M. Ljung. 2015. *Time series analysis: Forecasting and control*. Hoboken: Wiley.
- Breiman, L. 1996. Bagging predictors. *Machine Learning* 24: 123–140.
- Breiman, L. 2001. Random forests. *Machine Learning* 45: 5–32.
- Breiman, L., J.H. Friedman, R.A. Olshen, and C.J. Stone. 1984. *Classification and regression trees*. Statistics/probability series.
- Brouwer, G.D., and N.R. Ericsson. 1998. Modeling inflation in Australia. *Journal of Business & Economic Statistics* 16 (4): 433–449.
- CMIE. 2022. *Dataset of all variables*. Retrieved March 22, 2023 from CMIE Economic Outlook.
- Das, A., K. Lahiri, and Y. Zhao. 2019. Inflation expectations in India: Learning from household tendency surveys. *International Journal of Forecasting* 35 (3): 980–993.
- Garson, G.D. 1991. Interpreting neural-network connection weights. *AI Expert* 6 (4): 46–51.
- Goh, A.T. 1995. Back-propagation neural networks for modeling complex systems. *Artificial Intelligence in Engineering* 9 (3): 143–151.
- Gordon, R.J. 1975. The demand for and supply of inflation. *The Journal of Law and Economics* 18 (3): 807–836.
- Goulet Coulombe, P., M. Leroux, D. Stevanovic, and S. Surprenant. 2022. How is machine learning useful for macroeconomic forecasting? *Journal of Applied Econometrics* 37 (5): 920–964.
- Goyal, A., and P. Parab. 2019. Modeling heterogeneity and rationality of inflation expectations across Indian households. Indira Gandhi Institute of Development Research, Working Paper.
- Hall, R.E., and T.J. Sargent. 2018. Short-run and long-run effects of Milton Friedman's presidential address. *Journal of Economic Perspectives* 32 (1): 121–134.

- Hoerl, A.E., and R.W. Kennard. 1970. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics* 12 (1): 55–67.
- Inoue, A., and L. Kilian. 2008. How useful is bagging in forecasting economic time series? A case study of US consumer price inflation. *Journal of the American Statistical Association* 103 (482): 511–522.
- James, G., D. Witten, T. Hastie, and R. Tibshirani. 2013. *An introduction to statistical learning*, vol. 112, 18. New York: Springer.
- Liu, L.M. 1986. Identification of time series models in the presence of calendar variation. *International Journal of Forecasting* 2 (3): 357–372.
- Masini, R.P., M.C. Medeiros, and E.F. Mendes. 2023. Machine learning advances for time series forecasting. *Journal of Economic Surveys* 37 (1): 76–111.
- Medeiros, M.C., G.F. Vasconcelos, Á. Veiga, and E. Zilberman. 2021. Forecasting inflation in a data-rich environment: The benefits of machine learning methods. *Journal of Business & Economic Statistics* 39 (1): 98–119.
- Mullainathan, S., and J. Spiess. 2017. Machine learning: An applied econometric approach. *Journal of Economic Perspectives* 31 (2): 87–106.
- Nakamura, E. 2005. Inflation forecasting using a neural network. *Economics Letters* 86 (3): 373–378.
- Patra, M.D., and P. Ray. 2010. Inflation expectations and monetary policy in India: An empirical exploration. *IMF Working Paper*. <https://doi.org/10.5089/9781451982640.001>.
- Pradhan, R.P. 2011. Forecasting inflation in India: An application of ANN model. *International Journal of Asian Business and Information Management (IJABIM)* 2 (2): 64–73.
- Pratap, B., and S. Sengupta (2019). Macroeconomic forecasting in India: Does machine learning hold the key to better forecasts?
- Raj, J., S. Dhal, and R. Jain. 2008. *Imported inflation: The evidence from India*. New Delhi: Reserve Bank of India.
- Rani, S.J., V.V. Haragopal, and M.K. Reddy. 2017. Forecasting inflation rate of India using neural networks. *International Journal of Computer Applications* 158 (5): 45–48.
- Sharma, N.K., and M. Bicchal. 2018. The properties of inflation expectations: Evidence for India. *Economia* 19 (1): 74–89.
- Siami-Namini, S., N. Tavakoli, and A.S. Namin. 2019. A comparative analysis of forecasting financial time series using ARIMA, LSTM, and BiLSTM. arXiv preprint [arXiv:1911.09512](https://arxiv.org/abs/1911.09512).
- Sims, C.A. 1990. Macroeconomics and reality. *Econometrica: Journal of the Econometric Society* 1: 1–52. <https://doi.org/10.2307/1912017>.
- Stock, J.H., and M.W. Watson. 1999. Forecasting inflation. *Journal of Monetary Economics* 44 (2): 293–335.
- Stock, J.H., and M.W. Watson. 2002. Macroeconomic forecasting using diffusion indexes. *Journal of Business & Economic Statistics* 20 (2): 147–162.
- Stock, J.H., and M.W. Watson. 2007. Why has US inflation become harder to forecast? *Journal of Money, Credit and Banking* 39: 3–33.
- Thakur, G.S.M., R. Bhattacharyya, and S.S. Mondal. 2016. Artificial neural network based model for forecasting of inflation in India. *Fuzzy Information and Engineering* 8 (1): 87–100.
- Tibshirani, R. 1996. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (methodological)* 58 (1): 267–288.
- Zhang, G., B.E. Patuwo, and M.Y. Hu. 1998. Forecasting with artificial neural networks: The state of the art. *International Journal of Forecasting* 14 (1): 35–62.
- Zou, H. 2006. The adaptive lasso and its oracle properties. *Journal of the American Statistical Association* 101 (476): 1418–1429.
- Zou, H., and T. Hastie. 2005. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (statistical Methodology)* 67 (2): 301–320.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.